1 The Limit Comparison Test

1. Consider the improper integral

$$\int_1^\infty \frac{x}{x^2+\sqrt{x}+1}dx. \tag{1.1}$$

(a) For large values of x, approximate the integrand in (1.1) as a power of x. Based on this approximation, do you expect the integral to converge?

(b) Now let's try to prove your guess from part (a) using the Comparison Test. Show that

$$0 < \frac{x}{x^2+\sqrt{x}+1} < \frac{1}{x} \quad \text{for } x \geq 1. \tag{1.2}$$

(c) Using (1.2), what do you conclude about convergence of the improper integral in (1.1)?

(d) Suppose we could find some positive constant c such that

$$\frac{c}{x} \leq \frac{x}{x^2+\sqrt{x}+1}, \quad x \geq 1.$$

then what could you conclude about convergence of the improper integral in (1.1)?

The following result reduces the process of finding a suitable constant c to the computation of a single limit.

Theorem 1.3 (Limit Comparison Test)**.** *Suppose $f(x)$ and $g(x)$ are positive, continuous functions defined on $[a,\infty)$ such that*

$$\lim_{x\to\infty} \frac{f(x)}{g(x)} = c$$

for some positive real number c. Then $\int_a^\infty f(x)dx$ converges if and only if $\int_a^\infty g(x)dx$ converges.

2. From (1.2), $\frac{1}{x} > \frac{x}{x^2+\sqrt{x}+1} > 0$ when $x \geq 1$. Since the functions $f(x) = \frac{1}{x}$ and $g(x) = \frac{x}{x^2+\sqrt{x}+1}$ are also continuous on $[1,\infty)$, they satisfy the hypotheses of the Limit Comparison Test. Use the Limit Comparison Test to show that $\displaystyle\int_1^\infty \frac{1}{x^2+\sqrt{x}+1}dx$ indeed diverges.

3. Use the Limit Comparison Test to determine whether each of the following improper integrals converges or diverges.

   (a) $\displaystyle\int_1^\infty \frac{x^2+3x}{\sqrt{x^5+1}}\,dx$

   (b) $\displaystyle\int_1^\infty \sin\left(\frac{1}{x^2}\right)\,dx$ (*Hint:* Recall that $\lim_{t\to 0}\frac{\sin(t)}{t} = 1$)

A theorem similar to Theorem 1.3 holds for improper integrals of Type 2 where $\int_a^b f(x)dx$ has an infinite discontinuity at $c \in [a,b]$. In this case the relevant limit is either $\lim_{x\to b^-} \frac{f(x)}{g(x)}$ or $\lim_{x\to b^+} \frac{f(x)}{g(x)}$ (it doesn't matter which side we take the limit from, since we are just trying to determine convergence).

(c) $\displaystyle\int_1^2 \frac{x^2+x+1}{(x^2-1)^{1/3}}\,dx$ (*Hint*: What's important in this case is the degree to which the denominator goes to zero at 1. Since the denominator goes to zero as $(x-1)^{1/3}$, we take $g(x) = \frac{1}{(x-1)^{1/3}}$).

2 Proof of the Limit Comparison Test

2. We now prove Theorem 1.3.

Proof. (a) Show that there exists $x_0 \in [a, \infty)$ such that $|\frac{f(x)}{g(x)} - c| < \frac{c}{2}$ for all $x > x_0$.

(b) Show that the above inequality implies that, for $x > x_0$,

$$0 < \frac{c}{2}g(x) < f(x) < \frac{3c}{2}g(x). \tag{2.1}$$

(c) Apply the Comparison Test to (2.1) to show that $\int_{x_0}^{\infty} f(x)dx$ converges if and only if $\int_{x_0}^{\infty} g(x)dx$ converges.

(d) Conclude that $\int_a^{\infty} f(x)dx$ converges if and only if $\int_a^{\infty} g(x)dx$.

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